Homework 4

Mark Lee

November 23, 2025

Homework Deliverables

1 Single Load Level

Here are five snapshots of the helix centerline at different simulation times, 0s, 7s, 15s, 25s, and 35s.

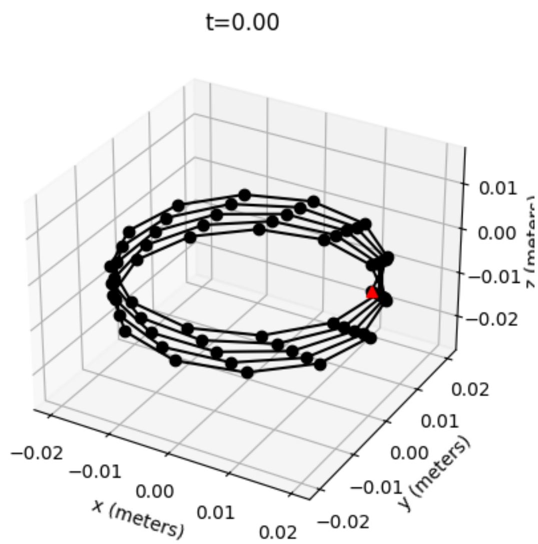

Figure 1: Helix configuration at starting position, t = 0.

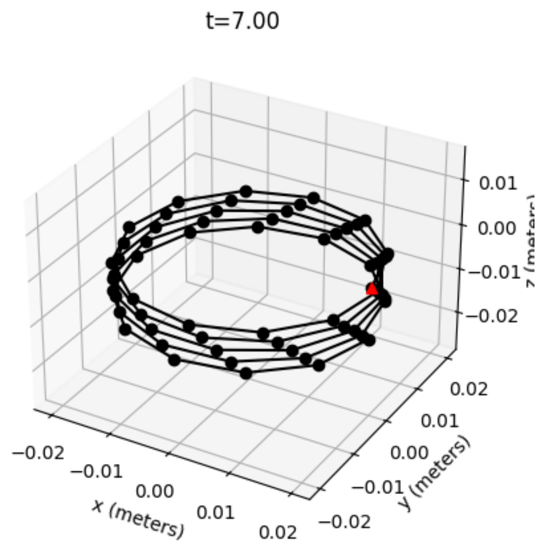

Figure 2: Helix configuration at t = 7

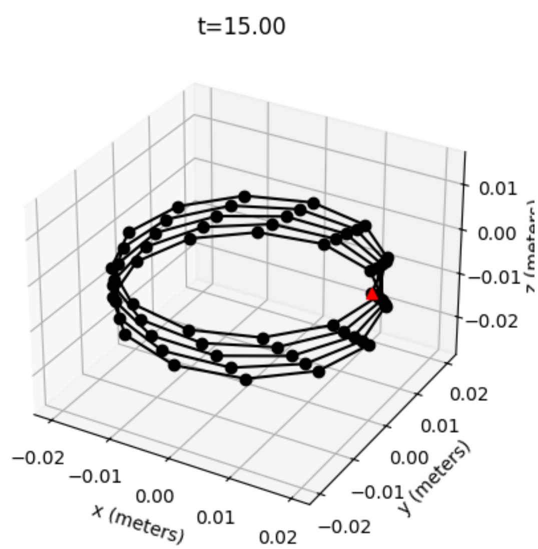

Figure 3: Helix configuration at t = 15

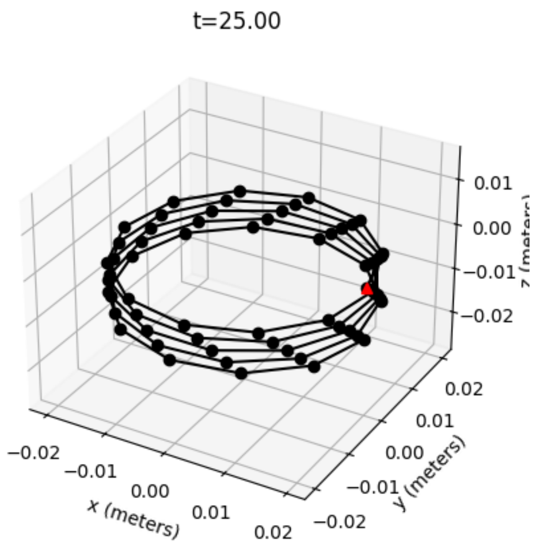

Figure 4: Helix configuration at t = 25

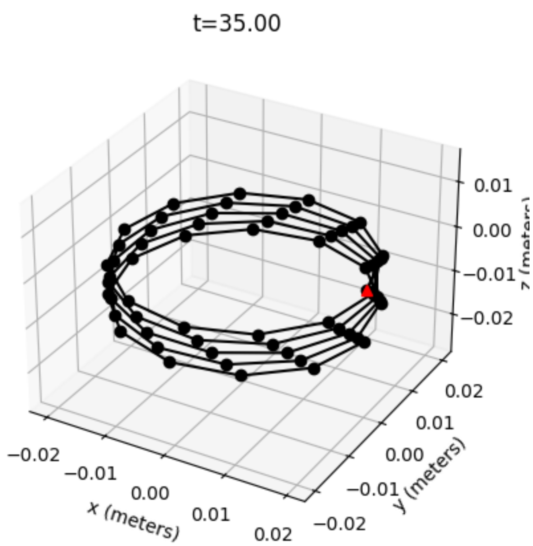

Figure 5: Helix configuration at t = 35

The system is considered to have reached steady state when the peak-to-peak oscillation amplitude over a 2-second window is less than 5% of the mean displacement.

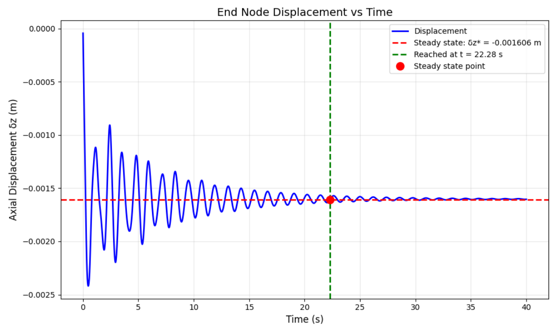

Figure 6: End Node Displacement (δz) vs Time

The characteristic force is $F_{\text{char}} = 0.0785$ N, and the system reaches steady state at $t = 22.8$ s with an axial displacement of $\delta_z^* = -0.001606$ m.

2 Force Sweep and Linear Fit

2.1 Methodology

To determine the linear stiffness of the helical spring, a force sweep was conducted by varying the applied axial force F logarithmically from $0.01F_{\text{char}}$ to $10F_{\text{char}}$, where $F_{\text{char}} = EI/L_{\text{axial}}^2$ is the characteristic force. A total of 15 force levels were tested using logarithmic spacing (`logspace` in NumPy). For each force level, the simulation was initialized from the undeformed helical configuration and run until steady state was reached according to the criterion established in Part 1 (peak-to-peak variation less than 5% of mean displacement over a 2-second window).

2.2 Linear Stiffness Extraction

The linear stiffness k was extracted from the steady-state displacement data using a zero-intercept linear fit. To ensure that only data from the linear regime were used for the fit, force levels satisfying $F < 3F_{\text{char}}$ were selected. The stiffness was determined using a least-squares fit with the intercept constrained to zero. This approach ensures that the fitted relationship passes through the origin, as required by the physics of the problem.

2.3 Results

The linear stiffness extracted from the simulation data is $k = 0.01$ N/m.

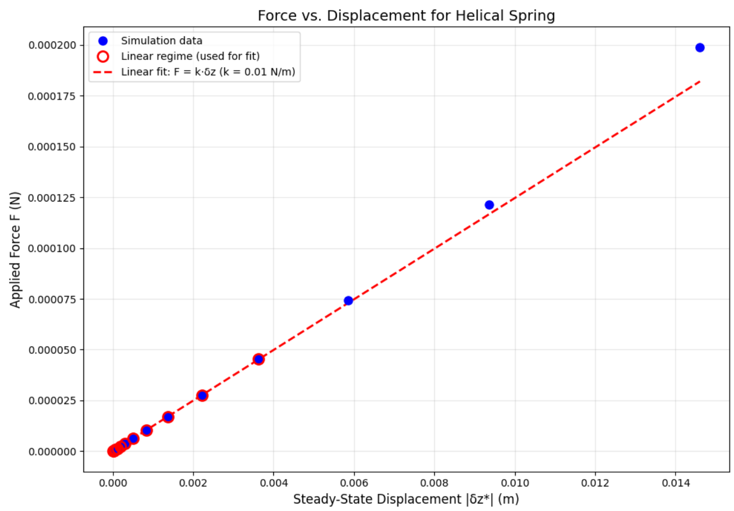

Figure 7: Applied force F vs steady-state displacement $|\delta_z^*|$. Blue circles represent all simulation data points, red circles with hollow centers indicate data used for the linear fit ($F < 3F_{\text{char}}$), and the red dashed line shows the zero-intercept linear fit $F = k\delta_z^*$.

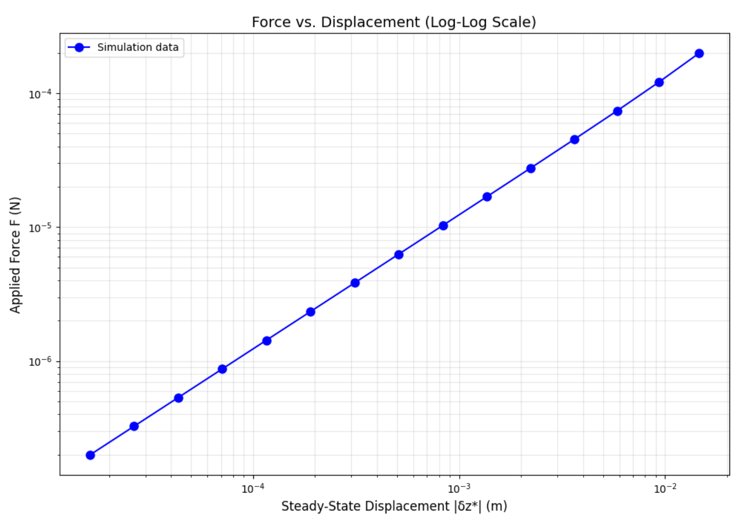

Figure 8: Applied force F vs steady-state displacement $|\delta_z^*|$ on a log-log scale, showing the full range of force levels tested. The nonlinear behavior at higher forces is evident from the deviation from a straight line.

At small forces, the response is approximately linear, while at larger forces the slope decreases, indicating stiffening or other nonlinear effects.

3 Diameter Sweep and Comparison with Textbook Prediction

3.1 Motivation and Theory

Textbooks often approximate the axial stiffness of a close-coiled helical spring using the classical formula:

$$k_{\text{text}} = \frac{Gd^4}{8ND^3} \tag{1}$$

where G is the shear modulus, d is the wire diameter, N is the number of turns, and D is the mean coil diameter. This formula is derived assuming that the spring deforms primarily through torsion of the wire, with negligible bending contributions, and is valid only in the linear, small-deflection regime.

In this part, we examine how the DER-based numerical stiffness compares to this classical prediction by varying the helix diameter D while keeping all other parameters fixed.

3.2 Methodology

The helix diameter D was varied from 0.01 m to 0.05 m using 10 equally spaced values. For each diameter, the helical geometry was reconstructed and a force sweep was conducted with 10 force levels ranging from $0.01F_{\text{char}}$ to $10.0F_{\text{char}}$, where $F_{\text{char}} = EI/L_{\text{axial}}^2$. The linear stiffness k was extracted using the zero-intercept linear fit method, and the textbook prediction k_{text} was calculated using Equation (1).

Numerical instabilities were encountered at higher force levels, particularly for larger diameter springs where large deformations violated small-deflection assumptions. Only 5–6 force levels per diameter (typically $F \leq 0.5F_{\text{char}}$) successfully converged. Stiffness values were extracted using only the successfully converged data points within the linear regime.

3.3 Results

The stiffness values extracted from simulations range from $k \approx 0.020$ N/m at $D = 0.05$ m to $k \approx 1.36$ N/m at $D = 0.01$ m, demonstrating the strong dependence of stiffness on coil diameter. All ten diameter values were successfully analyzed, with a mean stiffness ratio of $k_{\text{sim}}/k_{\text{text}} = 1.45 \pm 0.33$.

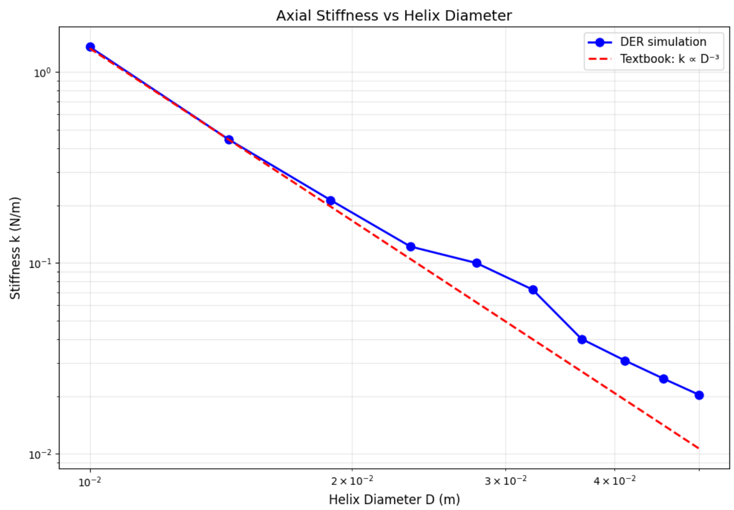

Figure 9: Comparison of DER simulation results with the textbook prediction. The horizontal axis represents the textbook parameter $Gd^4/(8ND^3)$ and the vertical axis shows the extracted stiffness k. Blue circles are simulation data, and the red dashed line with slope 1 represents the textbook relation (Equation (1)). Perfect agreement would result in all points lying on the red line.

This shows the primary comparison between simulation and theory. The data points lie consistently above the textbook line for most diameter values, indicating that the DER simulations predict higher stiffness than the classical formula.

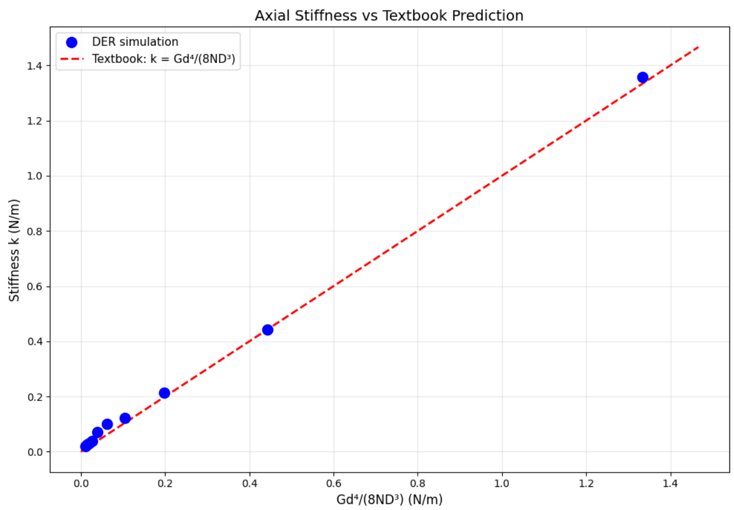

Figure 10: Axial stiffness k vs helix diameter D on a log-log scale. Blue circles with solid line represent DER simulation results, and the red dashed line shows the textbook prediction $k \propto D^{-3}$. Both curves exhibit the expected D^{-3} power-law dependence, with the simulation results consistently above the textbook prediction for larger diameters.

This plot clearly shows that both the simulation and textbook predictions follow a D^{-3} power law, as expected from the theoretical derivation.

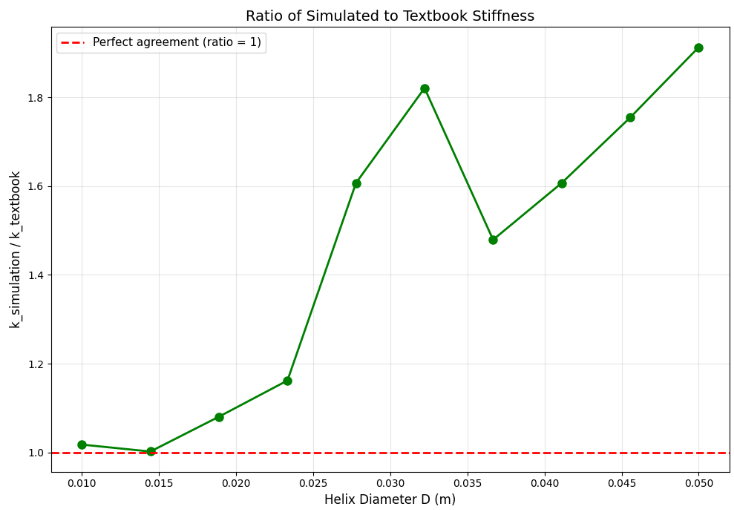

Figure 11: Ratio of simulated stiffness to textbook prediction ($k_{\text{sim}}/k_{\text{text}}$) vs helix diameter D. The red dashed line at ratio = 1 represents perfect agreement. Values above 1 indicate that simulations predict a stiffer spring than the textbook formula.

3.4 Discussion

The comparison reveals both qualitative agreement and quantitative differences between the DER simulations and the classical textbook formula. The textbook formula (Equation (1)) assumes that the spring deforms purely through torsion of the wire, neglecting bending effects. However, the DER model accounts for both bending and twisting energies, providing a more complete representation of the spring's mechanical behavior.

The results demonstrate that for close-coiled helical springs with pitch comparable to the wire diameter, the classical torsion-only formula may underestimate the actual stiffness, particularly for springs with larger coil diameters. This finding has practical implications for spring design applications where accurate stiffness predictions are critical.

References

[1] M. K. Jawed, "Lecture 13," Dept. of Mechanical and Aerospace Engineering, Univ. of California, Los Angeles, 2025. [Class handout].

[2] M. K. Jawed, "Lecture 12," Dept. of Mechanical and Aerospace Engineering, Univ. of California, Los Angeles, 2025. [Class resource].